

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2611–2616; original printed p. 88; physical source-PDF pp. 77–78; recomposed running pp. 69–70. Line 2617 is blank. The initial-term computation begins at line 2618 and is excluded. This unit has passed independent source/build/render review; cumulative integration remains pending.

One sees easily that $P^\bullet$ is the simple complex associated with $Q^{\bullet\bullet}$. Let us compute the abutment of the spectral sequence, that is, the homology of $\underline{E}P^\bullet$. To this end, using the fact that $L^\bullet$ is projective of finite type in every degree, one proves that

$$L^\bullet \cong \operatorname{Hom}^\bullet(L'^\bullet, A^\bullet).$$

From the homomorphism $a: A^\bullet \rightarrow CA^\bullet$, one obtains a homomorphism

$$b: \operatorname{Hom}^\bullet(L'^\bullet, A^\bullet) \longrightarrow \operatorname{Hom}^\bullet(L'^\bullet, CA^\bullet),$$

or equivalently a homomorphism

$$c: L^\bullet \longrightarrow P^\bullet. \tag{1.5}$$

This being said, it is easy to see, using the fact that $L'^\bullet$ is projective of finite type in every degree and bounded on the left, that (1.5) is an injective resolution of $L^\bullet$. Using Proposition 1.4, one concludes that $P^\bullet$ is homotopy equivalent to $I^\bullet$, where $I^\bullet$ is the injective resolution of M introduced above in (1.1). It follows that the *abutment* of the first spectral sequence of the double complex $\underline{E}Q^{\bullet\bullet}$, which is $H^*(\underline{E}P^\bullet)$, is *isomorphic* to $R^*\underline{F}(M)$.